SSL Log Investigation — Suspicious Certificate Issuer Analysis

Platform:	Elastic Stack (Kibana + Elasticsearch)
Index:	`ssl_log_analysis`
Date of Investigation:	April 7, 2024
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Investigation Overview

This investigation focused on identifying anomalous SSL/TLS certificate activity within network traffic logs. The goal was to detect certificates issued by **untrusted or suspicious Certificate Authorities (CAs)** that could indicate man-in-the-middle (MITM) attacks, malware command-and-control (C2) communication, or unauthorized certificate usage.

Query Used

A targeted Kibana query was constructed to isolate traffic associated with a suspicious certificate issuer:

```
issuer : "c0m0d0.com"
```

Findings

Field	Value
Timestamp	Apr 7, 2024 @ 22:07:36.000
Issuer	`c0m0d0.com`
Common Name (CN)	`life23.net`
Days Valid	126
Origin IP (id.orig_h)	`10.0.1.128`
Organization (org)	`life23.com`
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Analysis

1. **Issuer Impersonation:** The issuer `c0m0d0.com` is a deliberate misspelling of the legitimate Certificate Authority `comodo.com`. This resembles a classic typosquatting technique often used to make fraudulent certificates appear legitimate at a glance.
2. **Mismatched Organization Fields:** The certificate's Common Name (`life23.net`) and Organization (`life23.com`) do not correspond cleanly, which increases suspicion around the certificate's authenticity.
3. **Short Validity Period:** A validity period of only **126 days** may align with short-lived malicious infrastructure, where attackers rotate domains and certificates to reduce detection windows.
4. **Internal Origin IP:** The originating IP address `10.0.1.128` is an internal network address, suggesting that a device inside the environment communicated using this suspicious certificate.

Threat Indicators

Indicator	Type	Severity
`c0m0d0.com`	Suspicious Certificate Issuer	High
`life23.net`	Suspicious Domain (CN)	Medium
`10.0.1.128`	Internal Host — Potential Compromise	High

Recommended Actions

1. Block outbound traffic to domains associated with `c0m0d0.com` and `life23.net` at the firewall or proxy level.
2. Isolate the internal host at `10.0.1.128` for further forensic and endpoint investigation.
3. Alert the incident response team to determine whether additional internal systems communicated with the same issuer.
4. Update threat intelligence or detection content to include `c0m0d0.com` as a suspicious indicator.
5. Review SSL inspection and certificate validation controls to ensure suspicious issuers are detected promptly.

Tools Used

Tool	Purpose
Kibana	Log visualization and query interface
Elasticsearch	Backend log indexing and search engine

Conclusion

The SSL log investigation identified a suspicious certificate issued by `c0m0d0.com`. The combination of a deceptive issuer name, questionable domain details, and an internal originating IP address presents a credible signal of potentially malicious activity. Immediate containment and escalation are warranted.

Prepared for portfolio and cybersecurity documentation purposes.